

GC9A01 1.28" Round LCD

240 × 240 IPS · 4-wire SPI · GC9A01 controller

Round graphic display

240²

65K colours



A circular IPS LCD panel driven by the GC9A01 controller over a simple 4-wire SPI bus. Full 240 × 240 resolution edge-to-edge in a round format, wide IPS viewing angles, and an LED backlight. In this project it's the entire user interface — every radar page, card and weather readout is painted here.

DIAGONAL

1.28" round

RESOLUTION

240 × 240

PANEL

IPS full view

COLOUR

65K / 262K

INTERFACE

4-wire SPI

VOLTAGE

3.3 V

Panel & interface specification

DISPLAY

Driver IC

ELECTRICAL & BUS

Interface

Type	
Shape	Logic / supply
Resolution	
Active area	SPI clock
Colour depth	Module PCB
Backlight	Backlight pin
	Driver library

3.3 V only. VCC and all logic pins expect 3.3 V. Do not connect VCC to a 5 V rail.

Pinout (7-pin SPI header)

VCC 3.3 V power	GND ground	SCL SPI clock	SDA SPI data (MOSI)	RST reset	DC data / command	CS chip select
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Some modules add a separate **BLK/BL** backlight pin. If present, tie it to 3V3 (or a PWM GPIO for dimming). If absent, the backlight is always on.

As wired in the MUC Desk Radar

DISPLAY	ESP32-C3 PIN	SIGNAL
	3V3	Power (3.3 V)
	GND	Ground

GPI04	SPI clock
GPI03	SPI data
GPI010	Data/command
GPI01	Chip select
GPI00	Reset
3V3	Backlight on

Rendering note: the visible glass ends near radius 116 of 120, so the firmware keeps text and markers inside a circular safe area and clears in circles rather than rectangles — worth knowing if you customise the UI. If your panel shows inverted colours, flip the controller's invert setting in the display init.

GC9A01 1.28" 240×240 ROUND LCD · REFERENCE CARD · dimensions typical for common breakout modules